| Market Potential | | Sector |
|---|---|---|
| ₹11,000 cr | - | Agriculture |
| ₹13,000 cr | - | Infrastructure (Roads, Railways, Pipelines, Power, Telecom) |
| ₹3,000 cr | - | Mining |
| ₹10,000 cr | - | Defence and Homeland Security |
| ₹3,000 cr | - | Rural and Urban Development |
| ₹4,000 cr | - | Aerial Cinematography and Entertainment |
| ₹6,000 cr | - | Others (Inspection, Logistics, Healthcare, Insurance) |
| ₹50,000 cr | - | Total |

*Fig. 1: Market potential (five years) of the drone industry (Source: Primary Research by Drone Federation of India)*

to cover an acre in ten minutes, ensuring the operator's safety, conserving resources, and reducing labour needs.

***Military.*** Drones in logistics are revolutionising mid-mile deliveries, carrying 40-50kg of cargo between hubs. The Indian army is adopting drones for high-altitude deliveries, reducing dependence on helicopters, which face weight and air density challenges. Drones can double the payload by making helicopters autonomous and removing unnecessary equipment. This innovation highlights drones' potential in defence logistics and cargo.

***Mining.*** Drones offer a safer and more efficient way to conduct mine audits. Mining audits traditionally halt operations for two days, using manual surveyors who navigate risky terrains and record data on paper, which can be error prone. Drones, however, can survey the same area within a lunch break, capturing high-resolution photos to create a 3D digital model of the mine. This model offers precise measurements and, because data is geotagged, it's resistant to tampering, showcasing drones as a game-changer for mining audits.

***Renewable energy.*** For renewable energy, windmills and solar parks often rely on IoT sensors or robots for maintenance. Traditionally, inspecting windmills required personnel to climb and examine the blades, risking safety. Drones can perform these inspections quickly, safely, and with higher accuracy due to advanced camera technology, offering more insights than manual methods.

The market potential of the drone industry is depicted in Fig. 1.

From meticulous mapping of rural lands to aiding in maintaining renewable energy sources, drones are at the forefront of technological transformation. Drones are revolutionising data capture and analysis. Across India's 660,000 villages, drones record high-resolution imagery to create accurate land records. Traditionally, many village properties had unclear records, leading to disputes and encroachments. Drones provide precise digital land records, enabling landowners to leverage their property as a financial asset. This technology has a market potential of over 500 billion, spanning component and drone manufacturing, drone services, training, and software development. It opens doors for businesses in these sectors and can generate over 100,000 direct jobs, especially for drone pilots. And this isn't just about fancy flight manoeuvres; it's about data. High-resolution imagery, exact geographical mapping, creating digital twins, and providing analytical insights are where drones are changing the game.

## A push from the government

The journey of drone regulation in India has seen its ups and downs. It started with a blanket ban in 2014, primarily due to safety and security concerns. However, by 2021, after multiple revisions, Indian government unveiled a set of regulations that favoured the industry. A straightforward process was implemented, emphasising safety, security, and accountability. The new policies have made it as simple as launching a software startup. With benefits such as financial incentives and an emphasis on local production, India's stance is clear—to create, innovate, and lead the drone revolution for local markets and the world. The government recognised the potential benefits and, in sub-

*Agriculture drone spraying water fertiliser on the sunflower field*